Slide 1 — The question

How should I invest my calibration points?

The answer depends on what we are trying to learn.

Validation learns the method. Routine calibration checks that the validated method still works.

Slide 2 — Same budget, different information

With 15 measurements, we can choose:

- **15 x 1:** more information about response shape;
- **5 x 3:** a balance of levels and repeatability;
- **3 x 5:** precise repeatability estimates at low, middle, and high levels.

No arrangement is best for every question.

Slide 3 — First establish the mean response

Linearity is not established by a high R^2 or a nice-looking line.

During validation, use **many distributed concentration levels** to look for curvature, saturation, and regional lack of fit. Add replication so that normal scatter is not confused with a change in the mean response.

Slide 4 — Then establish the variance structure

A linear response can have non-constant variance.

Replicates reveal whether error is approximately:

- constant in absolute units;
- constant in relative terms; or
- mixed, with a low-level noise floor.

Weights are not a convention. They are a model of that variance.

Slide 5 — Use WLS only when its weights are credible

WLS gives more influence to measurements that are expected to be more precise.

It is useful only when the weights describe the method’s real variability. The variance pattern must therefore be estimated, checked, and monitored—not chosen because a familiar formula looks convenient.

Slide 6 — A practical routine design

After a linear mean and variance structure have been validated:

Use 3 x 5 with WLS estimated from replicate data

It is compact and auditable:

- low, middle, and high standards check the range;
- five replicates estimate local variability;
- data-estimated weights are preferable to a habitual $1/x^2$ rule.

Slide 7 — QC is the final check

Use independent QCs at relevant low, middle, and high concentrations.

Track:

- bias and relative prediction error;
- prediction-interval width; and
- **coverage**: does a nominal 95% interval contain about 95% of true QC concentrations?

If variance or QC prediction changes, investigate; do not silently retain the old weight rule.

Slide 8 — The answer

Invest broadly first, then monitor intelligently.

Validation: many concentration levels + replication to define linearity and variance.

Routine: 3 x 5 + data-estimated WLS + QC checks across the range.

This is how the calibration design, the statistical model, and the analytical decision all support the same workflow.

#AnalyticalChemistry #Chemometrics #Calibration #WeightedLeastSquares